Pergamon

Tetrahedron Letters 43 (2002) 5723–5726

TETRAHEDRON LETTERS

Unexpected S_N2'-type addition–elimination reactions of 1-aryl-2,3-allenols with LiX. Synthesis and synthetic application of 1-aryl-3-halo-1,3-dienes

Shengming Ma* and Guangwei Wang

State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 354 Fenglin Lu, Shanghai 200032, PR China

Received 23 April 2002; revised 13 June 2002; accepted 24 June 2002

Abstract—1-Aryl-3-halo-1,3-dienes were prepared from the sequential addition–elimination reaction of 1-aryl-2,3-allenols with LiX (X = Br, Cl) in HOAc in moderate to good yields. Here the aromatic substituent is crucial to this interesting transformation since no reaction was observed with 1-alkyl or perfluoroalkyl-2,3-allenols. The 1-aryl-3-halo-1,3-dienes prepared can be used as useful partners in Diels–Alder reactions with dienophiles leading to polycyclic quinone derivatives. © 2002 Elsevier Science Ltd. All rights reserved.

1,3-Dienes are important intermediates in organic synthesis,[1] especially as a partner in Diels–Alder reactions.[2] Recently, during our study of allene chemistry,[3] we observed that the Pd(0)-catalyzed coupling-cyclization of organic halides and 2,3-allenols afforded vinylic oxiranes[4] or 2,5-dihydrofurans,[5] whilst the Pd(0)-catalyzed reaction of 2,3-allenols with aryl halides in the presence of amines afforded α- or γ-aminoalcohols regioselectively and stereoselectively,[6] showing the synthetic potential of 2,3-allenols.[7–9] A recent study of Pd-catalyzed synthesis of (*Z*,*E*)-2-bromo-1,3-dienes from α-allenic acetates by Bäckvall et al.[10] prompted us to report an unexpected reaction of 1-aryl-2,3-allenols with lithium halides affording 1-aryl-3-halo-1,3-dienes.

Recently, Bäckvall et al. reported a palladium-catalyzed reaction of 4,5-allenols with LiBr in the presence of benzoquinone to afford vinylic tetrahydrofuran derivatives.[11] Based on this result we reasoned that the corresponding reaction of 2,3-allenols with lithium halides would produce 3-bromo-2,5-dihydrofurans. However, when a mixture of 1-phenyl-2,3-butadienol **1a**, LiBr (5 equiv.) and benzoquinone (2.5 equiv.) was treated with 5 mol% of $Pd(OAc)_2$ in HOAc, the reaction afforded an unexpected product whose structure upon X-ray diffraction analysis was determined to be that of **2a** (Fig. 1).

Using 5 mol% of $PdCl_2(CH_3CN)_2$ or $PdCl_2(PPh_3)_2$, the same product **2a** was formed, respectively. Surprisingly, it is interesting to note that this reaction can occur to afford **2a** even *in the absence of* any Pd catalyst. Here, it is obvious that the reaction proceeded via the intermediacy of 1-phenyl-3-bromo-1(*E*),3-butadiene **3a**.

With these results in hand, we studied the reaction in the absence of a dienophile in order to isolate the synthetically useful 2-halo-1,3-butadiene derivatives. When we ran this reaction with LiBr in a mixture of HOAc and CH_3CN (1:19), the reaction was complicated. Heating the mixture of $LiBr{\cdot}H_2O$ with **1a** in HOAc at 50°C afforded 1-phenyl-3-bromo-1,3-butadiene **3a** in 70% yield. The yield can be further improved to 76% when the reaction was carried out in the presence of 20 mol% of hydroquinone (Scheme 1).

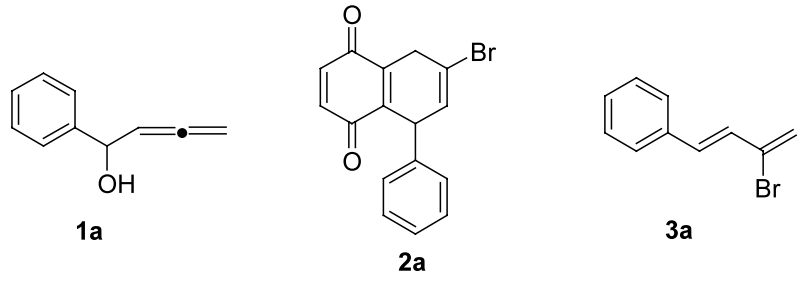
Thus, by applying these standard reaction conditions, the reactions of differently substituted 1-aryl-2,3-allenols with LiX in the presence of 20 mol% hydroquinone in HOAc afforded 1,3-butadienes. Some typical results are summarized in Table 1.

From the results in Table 1, it is obvious that the yields for the formation of 3-halo-1-aryl-1,3-butadienes are from moderate to good. The reaction was complete within a few hours. Both bromides and chlorides can be prepared. LiX monohydrate and anhydrous LiX did not show obvious differences in terms of yields of the products.

Keywords: allenes; alcohols; substitution; dienes.

* Corresponding author.

0040-4039/02/$ - see front matter © 2002 Elsevier Science Ltd. All rights reserved.
PII: S0040-4039(02)01207-8



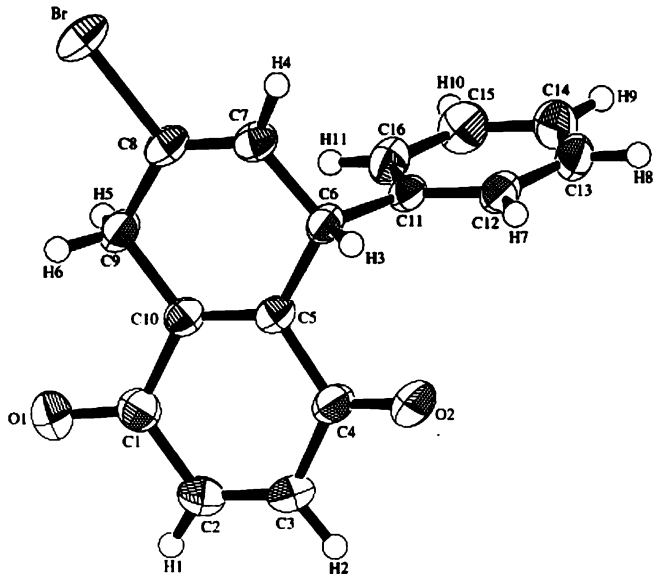
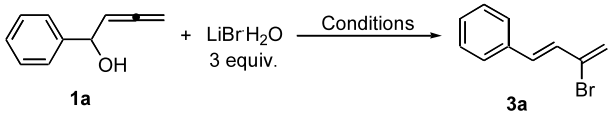


Figure 1. Structures **1a**, **2a** and **3a** and the ORTEP representation of **2a**.



AcOH/CH_3CN(1: 19)	reflux	10 h	complicated
AcOH	50°C	1 h	70 %
AcOH, HQ (20 mol%)	50°C	40 min.	76 %

Scheme 1.

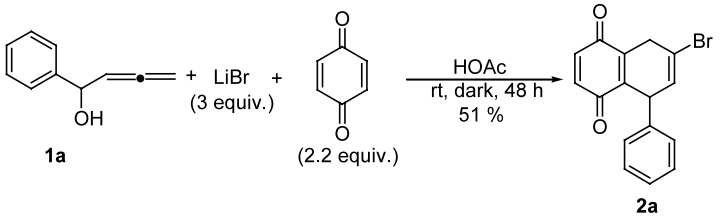
With 2-(*n*-butyl)-1-phenyl-2,3-butadienol **1e**, the reaction afforded *E*/*Z* mixtures of 3-halo-1-phenyl-2-(*n*-butyl)-1,3-butadienes (Scheme 2). The stereochemistry was determined by the ^1H–^1H NOESY spectra of **3j**.

This reaction can also be extended to the tertiary alcohol **1f** to afford the corresponding products in good yields (Scheme 3).

However, it is interesting to observe that the corresponding reaction of 1-alkyl or perfluoroalkyl-substituted allenols **1g** and **1h** did not yield the expected products (Scheme 4).

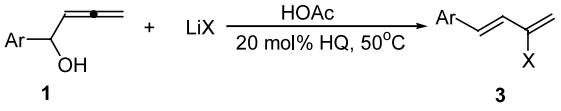
The reaction is believed to proceed through an S_N2'-type addition–elimination process.[12] In HOAc, the phenyl group may stabilize the corresponding carbocationic intermediate formed by the interaction of HOAc with the hydroxy group of 1-phenyl-2,3-allenols.

The dienes prepared can be used for the synthesis of polycyclic quinones via Diels–Alder oxidation reactions. 1,4-Naphthoquinones are a class of compounds with biological activities such as antiinflammatory, antitumor, antiparasitic and antimicrobial action.[13] Thus, we used benzoquinone as the dienophile to optimize the reaction conditions for the one-pot addition–elimination-Diels–Alder process. The reaction afforded the Diels–Alder adduct **2a** in 51% yield at rt in the dark (Eq. (1)). Using these standard reaction conditions, the

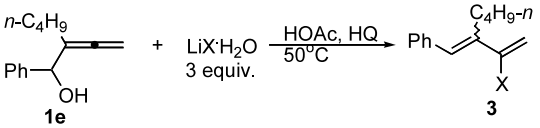


(1)

Table 1. Synthesis of 1-phenyl-3-halo-1,3-butadiene via the reaction of 1-phenyl-2,3-allenols with LiX in HOAc

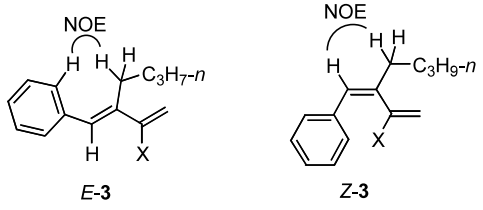


Entry	**1** (Ar)	LiX	Time (h)	Yield of **3** (%)
1	Ph (**1a**)	$LiBr{\cdot}H_2O$	40 min	76 (**3a**)
2	Ph (**1a**)	LiCl	2	60 (**3b**)
3	*p*-$CH_3OC_6H_4$ (**1b**)	LiBr	2	33 (**3c**)
4	*p*-$CH_3OC_6H_4$ (**1b**)	LiCl	2	53 (**3d**)
5	*o*-ClC_6H_4 (**1c**)	$LiBr{\cdot}H_2O$	1.5	82 (**3e**)
6	*o*-ClC_6H_4 (**1c**)	LiCl	1.5	60 (**3f**)
7	Naphthyl (**1d**)	$LiBr{\cdot}H_2O$	0.5	81 (**3g**)
8	Naphthyl (**1d**)	$LiCl{\cdot}H_2O$	0.5	88 (**3h**)

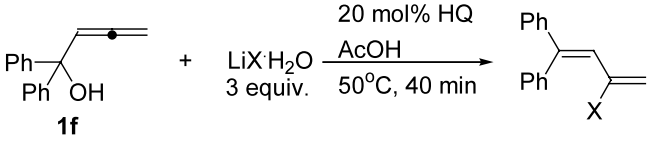


X = Br 3.0 h 87 % (*Z*/*E* = 1.4:1) (**3i**)

X = Cl 3.5 h 73 % (*E*/*Z* = 5.2:1) (**3j**)



Scheme 2.



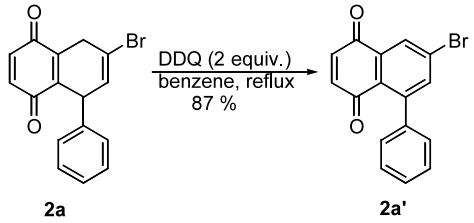
X = Br 40 min 90 % (**3k**)

X = Cl 40 min 85 % (**3l**)

Scheme 3.

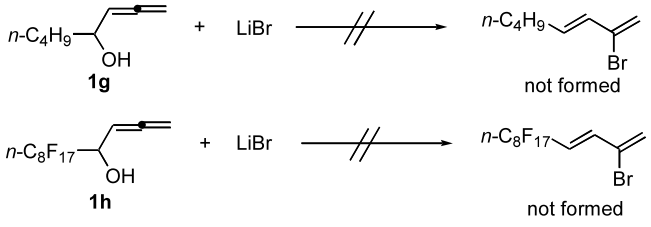
reaction of **1a**, LiBr and naphthoquinone afforded **2b** in 72% yield (Scheme 5).

Upon treatment with DDQ, **2a** was oxidized to afford the naphthoquinone derivative **2a**′ in 87% yield (Eq. (2)).

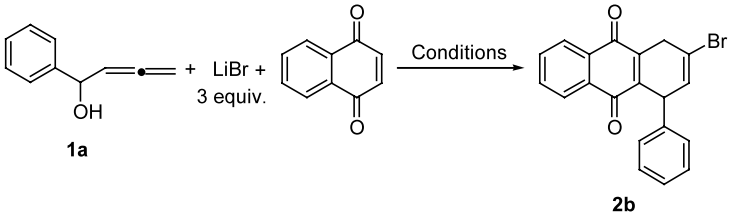


(2)

In conclusion, we have observed an interesting S_N2'-type addition–elimination reaction of 1-phenyl-2,3-allenols with LiX (X = Br, Cl). The 2-halo-1,3-dienes prepared show good reactivities towards quinones. Further studies on the scope and synthetic potentials of this reaction are being carried out in our laboratory.



Scheme 4.



Conditions	Yield(%)
AcOH, 40°C, 42 h	53
AcOH, dark, rt, 48 h	72

Scheme 5.

Acknowledgements

Financial support from the Major State Basic Research Development Program (Grant No. G2000077500), National Natural Science Foundation of China, and Shanghai Municipal Committee of Science and Technology are greatly appreciated. Shengming Ma is the recipient of the 1999 Qiu Shi Award for Young Scientific Workers issued by the Hong Kong Qiu Shi Foundation of Science and Technology (1999–2003).

References

1. For some of the most recent examples, see: (a) Qgasawara, M.; Ikeda, H.; Nagano, T.; Hayashi, T. *J. Am. Chem. Soc.* **2001**, *123*, 2089; (b) Abe, H.; Aoyagi, S.; Kibayashi, C. *J. Am. Chem. Soc.* **2000**, *122*, 4583.
2. (a) *Advances in Cycloaddition*; Curran, D. P., Ed.; JAI Press: Greenwich, 1993; Vol. 3; (b) Fringuelli, F.; Taticchi, A. *Dienes in the Diels–Alder Reaction*; John Wiley & Sons: New York, 1990.
3. For some of our recent results on the synthesis of cyclic compounds, see: (a) Ma, S.; Shi, Z. *J. Org. Chem.* **1998**, *63*, 6387; (b) Ma, S.; Shi, Z.; Wu, S. *Tetrahedron: Asymmetry* **2001**, *12*, 193; (c) Ma, S.; Wu, S. *J. Org. Chem.* **1999**, *64*, 9314. Ma, S. Wu, S. *Chem. Commun.* **2001**, 441; (d) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron Lett.* **1999**, *40*, 2393; (e) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron* **1999**, *55*, 12137; (f) Ma, S.; Zhao, S. *Org. Lett.* **2000**, *2*, 2495; (g) Ma, S.; Zhang, Z. *Chem. Commun.* **2000**, 117; (h) Ma, S.; Li, L. *Org. Lett.* **2000**, *2*, 941; (i) Ma, S.; Xie, H. *Org. Lett.* **2000**, *2*, 3801.
4. Ma, S.; Zhao, S. *J. Am. Chem. Soc.* **1999**, *121*, 7943.
5. Ma, S.; Gao, W. *Tetrahedron Lett.* **2000**, *41*, 8933.
6. Ma, S.; Zhao, S. *J. Am. Chem. Soc.* **2001**, *123*, 5578.
7. (a) For the Ru-catalyzed reaction of allenols in the presence of CO, see: Yoneda, E.; Kaneko, T.; Zhang, S.-W.; Onitsuka, K.; Takahashi, S. *Org. Lett.* **2000**, *2*, 441; (b) For the Ru-catalyzed cyclization of allenols with α,β-unsaturated enones, see: Trost, B. M.; Prinkerton, A. B. *J. Am. Chem. Soc.* **1999**, *121*, 10842.
8. For catalytic cycloisomerization of allenols, see: (a) Marshall, J. A.; Wang, X. *J. Org. Chem.* **1990**, *55*, 2995; (b) Marshall, J. A.; Wang, X. *J. Org. Chem.* **1991**, *56*, 960; (c) Olsson, L.-I.; Claesson, A. *Synthesis* **1979**, 743; (d) Nikam, S. S.; Chu, K.-H.; Wang, K. K. *J. Org. Chem.* **1986**, *51*, 745; (e) Gelin, R.; Gelin, S.; Albrand, M. *Bull. Soc. Chim. Fr.* **1972**, 1946; (f) Marshall, J. A.; Sehon, C. A. *J. Org. Chem.* **1995**, *60*, 5966; (g) Marshall, J. A.; Yu, R. H.; Perkins, J. F. *J. Org. Chem.* **1995**, *60*, 5550; (h) Hoffmann-Röder, A.; Krause, N. *Org. Lett.* **2001**, *3*, 2537.
9. For the Pd(0)-catalyzed coupling-cyclization reaction of allenols with aryl or vinyl halides, see: (a) Uemura, K.; Shiraishi, D.; Noziri, M.; Inoue, Y. *Bull. Chem. Soc. Jpn.* **1999**, 1063; (b) Kang, S.-K.; Baik, T.-G.; Kulak, A. N. *Synlett* **1999**, 324; (c) Kang, S.-K.; Yamaguchi, T.; Pyun, S.-J.; Lee, Y.-T.; Baik, T.-G. *Tetrahedron Lett.* **1998**, *39*, 2127; (d) Walkup, R. D.; Guan, L.; Kim, Y. S.; Kim, S. W. *Tetrahedron Lett.* **1995**, *36*, 3805; (e) Walkup, R. D.; Guan, L.; Mosher, M. D.; Kim, S. W.; Kim, Y. S. *Synlett* **1993**, 88.
10. Horváth, A.; Bäckvall, J.-E. *J. Org. Chem.* **2001**, *66*, 8120.
11. Jonasson, C.; Horváth, A.; Bäckvall, J.-E. *J. Am. Chem. Soc.* **2000**, *122*, 9600.
12. For hydrohalogenation reactions of electron-deficient allenes, see: (a) Ma, S.; Shi, Z.; Li, L. *J. Org. Chem.* **1998**, *63*, 4522; (b) Ma, S.; Wang, G. *Chin. J. Chem.* **1999**, *17*, 545; (c) Ma, S.; Li, L.; Xie, H. *J. Org. Chem.* **1999**, *64*, 5325; (d) Ma, S.; Wei, Q. *Eur. J. Org. Chem.* **2000**, 1939; (e) Ma, S.; Xie, H.; Wang, G.; Zhang, J.; Shi, Z. *Synthesis* **2001**, 713; (f) Ma, S.; Li, L.; Wei, Q.; Xie, H.; Wang, G.; Shi, Z.; Zhang, J. *Pure Appl. Chem.* **2000**, *72*, 1739. For an account, see: Ma, S.; Li, L. *Synlett* **2001**, 1206.
13. Dessolin, J.; Biot, C.; Davioud-Charvet, E. *J. Org. Chem.* **2001**, *66*, 5616.